IBC User Guide

Version 3.25.1 — Linux edition

Important notes

Auto-restart. Starting with version 3.15.0, IBC allows TWS/Gateway to use the auto-restart mechanism originally introduced in TWS 974/975. You can therefore run TWS/Gateway all week with a single login at the start of the week, under IBC's control. Configure this with the `AutoRestart` setting in the *Lock and Exit* section of the TWS/Gateway configuration dialog, or with `AutoRestartTime` in `config.ini`.

Auto-restart can lead to confusion if you sometimes want to start TWS/Gateway without IBC: see How to run TWS/Gateway without IBC.

Scope. Read Scope of this User Guide before going further: this project distributes a Linux edition only.

Introduction

Overview of IBC

IBC enables Interactive Brokers' Trader Workstation (TWS) and IB Gateway to be run in 'hands-free' mode, so that a user need not be present. This makes possible the deployment of unattended automated trading systems.

IBC loads TWS or the Gateway and then 'listens' for various events (such as the display of dialogs) that would normally require user intervention, and takes the appropriate action on the user's behalf. For example, it automates the login by filling the login dialog with your credentials and 'clicking' the login button.

Here are some of the things IBC does for you:

- starts TWS or the Gateway;
- logs you into TWS or the Gateway;
- clicks the YES button if the "Accept incoming connection?" dialog is displayed;
- allows TWS or the Gateway to restart each day without repeated authentication: authentication is only required the first time during the week that TWS or the Gateway run after 01:00 ET on Sunday;
- completes two-factor authentication, either by submitting a passcode it computes from a shared TOTP secret (`Ib2FA`), or by letting you acknowledge the alert in the IBKR Mobile app and retrying the login if it times out.

IBC also responds to commands sent to it by another program, for example to tell TWS/Gateway to shut itself down cleanly, to restart, or to reconnect.

Scope of this User Guide

This User Guide is intended to help you get started with IBC. It does not cover every feature in depth.

The configuration file `config.ini` that governs IBC's behaviour contains extensive notes on the various settings; it is the reference for anything not explained here.

This edition covers the **Linux distribution only**: the project publishes a single ZIP whose start scripts are Bash scripts. Windows and macOS editions are no longer produced, the upstream project `IbcAlpha/IBC` having been retired on 1 September 2026 and its repository archived read-only.

Getting Started

Checklist

Here is a summary of the steps you need to perform to get IBC up and running properly.

1. Install the **offline** edition of Trader Workstation or IB Gateway, and make sure it uses the English language setting. IBC does **not** work with the self-updating edition.
2. Install `xterm` if it is not already installed (to check, enter the command `xterm` in a terminal session). The start scripts open their diagnostic window in xterm unless you pass `-inline` as the first argument.
3. Download the IBC distribution ZIP, `IBCLinux-3.25.1.zip` (the name carries the version), from the releases page of the project (see Where to get IBC).
4. Install IBC (see Installing IBC). If you already have an existing installation, rename its folder before installing the new version in case you need to revert to it later.
5. Create a folder called `ibc` in your home directory, encrypted if your distribution supports it, to hold your configuration (see Password Security).
6. Copy `config.ini` from the IBC installation folder into that folder.
7. Edit `~/ibc/config.ini` with a text editor and set your credentials in `IbLoginId` and `IbPassword`. Set `Ib2FA` to your Base32 two-factor secret, or leave it empty to authenticate with a push notification or a manually entered code. It is advisable to use your paper-trading credentials at first, and for that you also need to set `TradingMode` to `paper`.
8. Check that the major version number of your TWS/Gateway installation is set in the start script you intend to use, `gatewaystart.sh` or `twsstart.sh`. To find it, first run TWS or the Gateway manually using the IBKR-provided icon, then click *Help > About Trader Workstation* or *Help > About IB Gateway*. You will see a line similar to this:

   ```
   Build 10.50.1f, Oct 5, 2026 4:18:53 PM
   ```

   The `TWS_MAJOR_VRSN` variable must then be set to `1050`, that is the first two parts of the version number with the period removed. The supplied scripts default to `1050`.
9. At this stage everything is set up to run IBC with its default settings, which start the Gateway (or TWS) and attempt to log in to your account. Run the relevant script from the IBC installation folder, for example `/opt/ibc/gatewaystart.sh`. If everything is satisfactory, shut down IBC by closing TWS/Gateway in the usual way.

   Information about the startup process is written to a log file, whose location is printed at startup (by default in `~/ibc/logs`). Include that file when reporting a problem with IBC.
10. Now edit `config.ini` further as needed: see Configuring IBC.
11. If you did not install TWS/Gateway and IBC in their default locations, or store the configuration file in the recommended location, edit the variables at the top of the start scripts accordingly. The scripts are commented to help you.
12. If you intend to run API programs that connect to TWS/Gateway, you must also edit the API settings in the TWS Global Configuration dialog.
13. If you want TWS/Gateway to restart every day during the week without you re-authenticating, set `AutoRestart` in the *Lock and Exit* section of the Global Configuration dialog, or `AutoRestartTime`

in `config.ini`. The only alternative is auto-logoff, which shuts TWS/Gateway down completely at the specified time, leaving you to restart it and re-authenticate.

Prerequisites

Java Runtime

IBC runs inside the same Java virtual machine as TWS/Gateway, so the Java runtime it uses is the one shipped with your TWS/Gateway installation. The start scripts pick that runtime from the installation automatically, which means you do not have to install Java yourself.

`IBC.jar` is compiled for Java 25 (class file version 69), so the runtime must be Java 25 or later. If your installation ships an older runtime, install Java 25 and point the `JAVA_PATH` variable in the start script at it; that variable overrides the runtime taken from the installation.

Interactive Brokers TWS or Gateway

Download and install the offline edition of Trader Workstation or IB Gateway from the Interactive Brokers website. The location of the downloads page varies from time to time and from country to country; at the time of writing it is reachable through the *Trading* menu, then *Platforms*, then *Download TWS*.

IBKR provides two modes of operation for TWS:

- an online, or self-updating TWS that receives updates automatically. IBC does not work with the self-updating TWS, so do not install that version;
- an offline or standalone TWS that never changes until you install another version. You must use this one with IBC.

The TWS installation contains the code for both TWS and the Gateway. The Gateway-specific downloads on IBKR's website contain the same code but install in a different place; either is fine. When you start IBC in Gateway mode it loads the Gateway installation, and in TWS mode the TWS installation.

It is safest to use the 'stable' offline version rather than the 'latest' for live trading: the latter is more likely to have bugs.

IBC needs TWS to operate in English so that it can recognise the dialogs it interacts with. Set the language by starting TWS manually (that is, without IBC) and selecting it on the initial login dialog; TWS remembers the setting when you subsequently start it using IBC.

Graphical session

IBC drives the real Swing user interface of TWS/Gateway, so it needs a display. On a desktop that is the session you are logged into. On a headless server, run an X server such as `Xvfb` and point `DISPLAY` at it before starting IBC.

Where to get IBC

IBC is distributed as a ZIP file containing the compiled program and the files listed below.

The latest version, and earlier ones, are published on the releases page of https://github.com/hartza-capital/IBC. That repository is the maintained source of IBC since the upstream project `IbcAlpha/IBC` was retired on 1 September 2026 and archived read-only.

The distribution ZIP, `IBCLinux-3.25.1.zip`, contains:

- the licence text, `COPYING.txt`;
- `IBC.jar`, the compiled Java code;
- a sample configuration file, `config.ini`;

- the start scripts `twsstart.sh` and `gatewaystart.sh`, plus the `scripts` sub-folder they call;
- command client scripts: `commandsend.sh`, `stop.sh`, `restart.sh`, `pause.sh`, `enableapi.sh`, `reconnectdata.sh`, `reconnectaccount.sh`;
- a text file called `version` containing the IBC version number;
- `README.txt`, which points at the project;
- this User Guide, `userguide.pdf`.

Source code and build scripts are not part of the ZIP; they are in the repository above.

Installing IBC

Installing IBC is just a matter of extracting the ZIP file to wherever you want to install it:

```
sudo unzip ~/Downloads/IBCLinux-3.25.1.zip -d /opt/ibc
cd /opt/ibc
sudo chmod o+x *.sh scripts/*.sh
```

`/opt/ibc` is the default location assumed by the scripts; using it minimises customisation. If you already have a previous installation, rename its folder (for example to `/opt/ibc.old`) so that you can refer back to your customisations.

Default Paths

The supplied scripts assume the defaults shown below, where `<username>` is your operating system user name, not your IBKR login id.

Item	Path
IBKR TWS/Gateway	`/home/<username>/Jts`
IBC program files	`/opt/ibc`
`config.ini`	`/home/<username>/ibc`
log files	`/home/<username>/ibc/logs`

If you store any of these items elsewhere, edit the variables at the top of the start scripts to reflect the new locations.

Password Security

To log in to TWS or IB Gateway, IBC needs your Interactive Brokers username and password. Secure them carefully to prevent unauthorised use by third parties.

IBC accepts the credentials in one of two ways:

- through the configuration file `config.ini`: this is the preferred method, because the file can be protected by the operating system;
- as command-line parameters when IBC is started: this method is strongly deprecated, because the command line of a process is readable by other users of the machine (for example through `ps`).

Protecting the configuration file

Store `config.ini` where other users of the machine cannot read it: within your home directory, in its own `ibc` folder. Then restrict the permissions:

```
chmod 700 ~/ibc
chmod 600 ~/ibc/config.ini
```

Consider encrypting the folder as well, so that another user with administrator privileges cannot read the contents even if they gain access to the files. On Linux the usual options are an encrypted home directory, a LUKS volume mounted under your home directory, or a per-directory tool such as `gocryptfs` or `encfs`.

Note that `Ib2FA` is as sensitive as the password itself: anyone holding the secret can generate valid passcodes.

Configuring IBC

IBC must be supplied with a configuration file. A specimen `config.ini` is included in the distribution ZIP; the scripts look for `~/ibc/config.ini` unless you point `IBC_INI` somewhere else.

`config.ini` contains detailed comments on every setting, and most of them have sensible defaults or are only needed in special situations. Here are the settings you are most likely to change:

`IbLoginId`
: Your IBKR username.

`IbPassword`
: Your IBKR password.

`Ib2FA`
: Base32 TOTP secret (the text IBKR shows alongside the QR code), used to complete two-factor authentication without any interaction. Leave it empty to use a push notification or a manual code. A value that is not valid Base32 stops IBC at startup.

`SecondFactorDevice`
: When several 2FA devices are registered, the item to select in the device list, spelled exactly as it appears there.

`TradingMode`
: Set to `paper` to use the paper-trading account, `live` otherwise; `live` is the default if the setting is absent.

`AcceptNonBrokerageAccountWarning`
: Logging in to a paper-trading account raises a dialog confirming that it is not a brokerage account, and TWS blocks API connections until it is accepted. `yes` (the default) accepts it automatically; `no` leaves it to you.

`AcceptIncomingConnectionAction`
: It is safest to set this to `reject` and to list the permitted API addresses in the TWS/Gateway configuration dialog instead.

`AutoRestartTime`
: Time at which TWS/Gateway restarts each day without re-authentication (`<hh:mm>`).

`ClosedownAt`
: Time at which IBC shuts TWS/Gateway down tidily, with no restart: `<hh:mm>` daily, or `<dayOfWeek hh:mm>` weekly, for example `Friday 22:00`.

`CommandServerPort`
: TCP port of the command server (7462 by convention); `0`, the default, disables it. Must differ between instances running on the same machine.

`ExistingSessionDetectedAction`
: What to do when the same IBKR user is already logged in elsewhere: `manual`, `primary`, `secondary` or `primaryoverride`.

`OverrideTwsApiPort`
: Forces the API listening port, whatever the value stored in TWS.

`IbDir` still exists, but using it is deprecated: auto-restart does not work when TWS settings are redirected that way. Set `TWS_SETTINGS_PATH` in the start script instead.

There are two ways IBC locates the configuration file:

- the simplest is to tell it in the start script, through the `IBC_INI` variable. This is the recommended

approach (the supplied scripts do exactly that) and lets you give the file any name you like;
- if you do not specify a file, IBC expects `config.ini` in `~/ibc`.

Starting IBC

The normal way to start IBC is to run one of the start scripts shipped in the ZIP: `gatewaystart.sh` for the Gateway, `twsstart.sh` for TWS. Due to their complexity you are strongly advised to use them rather than write your own.

Run the script from a terminal, or from any launcher that gives it a graphical session. Three things govern how it behaves:

- without arguments, it opens an xterm window showing the startup diagnostics and then logs in;
- with `-inline` as the first argument, the diagnostics are printed in the current terminal instead, and no xterm is used. This is the mode to use from `cron` or `systemd`;
- `HIDE`, when set to `YES` or `TRUE`, starts the diagnostic xterm window iconified. It has no effect with `-inline`.

The variables at the top of the script are the only part you may need to change:

`TWS_MAJOR_VRSN`
: Installed TWS/Gateway version, for example `1050` for build `10.50.1f`.

`IBC_INI`
: Location of `config.ini`; defaults to `~/ibc/config.ini`.

`TRADING_MODE`
: `live` or `paper`; overrides `TradingMode` in `config.ini`.

`TWOFA_TIMEOUT_ACTION`
: `exit` or `restart`: what to do when two-factor authentication times out (see Second Factor Authentication).

`IBC_PATH`
: IBC installation folder; defaults to `/opt/ibc`.

`TWS_PATH`
: TWS/Gateway installation folder; defaults to `~/Jts`.

`TWS_SETTINGS_PATH`
: Folder where TWS/Gateway stores its settings; defaults to `TWS_PATH`.

`LOG_PATH`
: Folder for the diagnostic log; defaults to `~/ibc/logs`. Unset, the log goes to the terminal; removed, nothing is logged.

`TWSUSERID`, `TWSPASSWORD`
: Credentials passed on the command line. Do not use them: put the credentials in `config.ini` instead.

`FIXUSERID`, `FIXPASSWORD`
: FIX credentials, Gateway only. Deprecated for the same reason.

`JAVA_PATH`
: Java installation to run TWS/Gateway and IBC; defaults to the runtime shipped with the installation.

`HIDE`
: `YES` or `TRUE` iconifies the diagnostic window.

The scripts refuse to start a second instance that uses the same `config.ini` (detected by looking for a `java` process carrying that path). That is what makes the periodic restarts described in Running IBC automatically safe.

Second Factor Authentication

You can use your mobile phone or tablet running Android or iOS to provide second factor authentication for your TWS login. To do that, install the IBKR Mobile app on the device and register it for second fac-

tor authentication using the button the app displays.

Once registered, every login to TWS or Gateway (including when IBC performs it) sends an alert to your device. Acknowledging the alert completes the login.

Automatic passcodes (TOTP)

IBC can also complete two-factor authentication with no interaction at all, by computing the passcode itself. Set `Ib2FA` in `config.ini` to the Base32 secret that IBKR shows next to the QR code when you register an authenticator:

```
Ib2FA=JBSWY3DPEHPK3PXP
```

IBC then derives the current six-digit passcode (RFC 4226/6238, HMAC-SHA1) and submits it on the 2FA dialog.

Two caveats:

- any non-empty value is turned into a passcode and submitted, so a placeholder such as `changeme` produces a wrong code that spends one of the few failed login attempts IBKR allows before locking the account. Leave `Ib2FA` empty until you have the real secret;
- a value that is not valid Base32 stops IBC at startup with an `INVALID_SETTING_VALUE` error, rather than letting it attempt a login.

If several 2FA devices are registered, `SecondFactorDevice` selects the one to use, spelled exactly as the device list shows it.

When the login does not complete

If you use the IBKR Mobile app and fail to acknowledge the alert within a fixed period (currently 3 minutes), the login cannot complete without manual intervention at TWS/Gateway. IBC can be told to detect this timeout and re-initiate the login, giving you another alert:

```
ReloginAfterSecondFactorAuthenticationTimeout=yes
```

The timeout/relogin mechanism can repeat any number of times until you acknowledge an alert and the login succeeds.

In some circumstances the login does not complete even though you acknowledged the alert. IBC can then shut down and restart, which repeats the login sequence and gives you another chance to receive an alert. This is controlled by `SecondFactorAuthenticationExitInterval`, the number of seconds IBC waits for the login to complete after you acknowledged the alert. For the automatic restart you must also set `TWOFA_TIMEOUT_ACTION=restart` in the start script.

If you have another mechanism that restarts IBC when it exits (for example `cron` or `systemd`), set `TWOFA_TIMEOUT_ACTION=exit` instead, so that both mechanisms do not react at the same time.

Running IBC automatically

crontab

On Linux you can use `crontab` to run `gatewaystart.sh` or `twsstart.sh` automatically. For example, to start the Gateway at 08:00 on Mondays:

```
0 8 * * 1 export DISPLAY=:10 && /bin/bash /opt/ibc/gatewaystart.sh -inline
```

The value of `DISPLAY` depends on how your system is configured.

Because the script exits immediately when an instance already runs with the same `config.ini`, you can do better than a single start: attempt to start IBC periodically and let it succeed only when it is not running.

```
0,15,30,45 * * * 1-5 export DISPLAY=:10 && /bin/bash /opt/ibc/gatewaystart.sh -inline
```

This restarts TWS/Gateway after an unexpected shutdown and, with `primaryoverride` as the `ExistingSessionDetectedAction`, also after a session was terminated from the IBKR Mobile app or the Client Portal.

systemd

A user unit does the same job with logging handled by the journal. Create `ibc-gateway.service` in `~/.config/systemd/user`:

```
[Unit]
Description=IBC + IB Gateway
After=graphical-session.target

[Service]
Type=simple
Environment=DISPLAY=:0
ExecStart=/opt/ibc/gatewaystart.sh -inline
Restart=on-failure
RestartSec=60

[Install]
WantedBy=default.target
```

Then enable it and make the user manager start it at boot:

```
systemctl --user daemon-reload
systemctl --user enable --now ibc-gateway
loginctl enable-linger "$USER"
```

`systemctl --user stop ibc-gateway` ends the session, whereas the `stop.sh` script lets IBC close TWS/Gateway tidily.

Multiple IBC instances

You may want to run more than one instance of TWS or the Gateway on the same computer, perhaps simultaneously. Common reasons:

- you want to run both your live and paper-trading accounts, especially if you need market data from both (IBKR only allows that when both instances run on the same computer);
- you have several logins for your live account;
- you trade on behalf of others, each with their own account;
- you want to trade in different regions at different times;
- you want to test a new TWS version on paper while your live account stays on the previous one.

TWS stores a large number of settings in a folder structure inside the installation folder, one sub-folder per username. Some files it creates while running are not separated by username and only one instance can hold them at a time, so multiple instances work as long as each logs in to a different username.

`TWS_SETTINGS_PATH` tells TWS/Gateway where to store its settings, so running instances simultaneously means giving each of them its own settings folder. You do not need to copy the TWS `.jar` files: all instances can load the same installation.

For example, to keep a live and a paper instance completely separate:

- install TWS in the default location, `~/Jts`;
- create two settings folders, `~/JtsLive` and `~/JtsPaper`;

- create two configuration files, `configLive.ini` and `configPaper.ini`, with the matching credentials in `IbLoginId` and `IbPassword` (plus `TradingMode=paper` in the paper one);
- copy `gatewaystart.sh` to `gatewaystartLive.sh` and `gatewaystartPaper.sh`;
- in each copy, set `IBC_INI` to its configuration file and `TWS_SETTINGS_PATH` to the matching settings folder, and give the two instances different `LOG_PATH` folders, for example `~/ibc/LiveLogs` and `~/ibc/PaperLogs`.

Running the two scripts then starts two instances, each logged in to a different account with its settings in its own folder. Do not let them share a log folder: they would write to the same log file and one of them would fail.

Using different TWS versions simultaneously

To use more than one version of TWS (for example to test a new one with your paper account while the live account stays on the previous version), install the versions in the normal way: TWS's installer places each one in its own folder named after the version number.

Then follow the advice in the previous section and make sure each start script has the `TWS_MAJOR_VRSN` of the version it is to load.

How to run TWS/Gateway without IBC when IBC is installed

For auto-restart to work properly, the IBC start script renames the TWS/Gateway executables by appending a `1` to the filename: `tws` becomes `tws1` and `ibgateway` becomes `ibgateway1`. If those files have their original names when auto-restart occurs, TWS/Gateway do restart, but without IBC, and every benefit of IBC is lost. For that reason, do not rename them back while IBC is running.

The scripts do not restore the original names when IBC exits, which can be confusing if you then want to start TWS/Gateway without IBC. Some options:

- rename the executables back to their original names before running them;
- run the renamed executables directly, for example `~/Jts/1050/ibgateway1`;
- edit the IB-supplied desktop launcher, or create a second one, to point at the renamed executable;
- if you have a script that starts TWS/Gateway without IBC, change it to use whichever of the two names currently exists;
- install a second copy of TWS/Gateway in another folder and use only that one without IBC, keeping the same settings through `TWS_SETTINGS_PATH`.

Command Server

IBC incorporates a command server that lets external programs influence its operation. To issue a command, first open a TCP connection to the port given by `CommandServerPort` in `config.ini`; the server is not started when that setting is `0`.

The client then sends the command as plain text and may read the socket for the answer: a line starting with `OK` when the command was accepted, `ERROR` otherwise, each followed by a message. A client may send several consecutive commands, and should finish with `EXIT` (which is not necessary after `STOP`, since that closes the socket).

By default the server accepts connections from the local machine only; list other permitted addresses or host names in `ControlFrom`, and restrict the listening interface with `BindAddress` on a multi-homed host. `CommandPrompt` sets a string printed after the connection is opened and after each command, which makes interactive use easier.

A simple way to use the command server is the `telnet` command. The distribution ZIP contains a script per command: `stop.sh`, `restart.sh`, `pause.sh`, `enableapi.sh`, `reconnectdata.sh`,

`reconnectaccount.sh`, all of which call `commandsend.sh`. Edit `commandsend.sh` first to set the address and port.

`STOP`
: Shuts TWS/Gateway down tidily, as the *File > Exit* menu command would.

`RESTART`
: Restarts TWS/Gateway as if the auto-restart time had arrived. Credentials from the current session are reused, so no second factor is needed. It cannot bypass IBKR's requirement that the session be shut down completely at some point on Sunday.

`PAUSE`
: Restarts TWS/Gateway like `RESTART`, but IBC then exits instead of relaunching it. The session is kept, so starting IBC again continues it without a new authentication; that is how a session is parked or moved between runs.

`ENABLEAPI`
: Ensures the *Enable ActiveX and Socket Clients* checkbox is set. TWS only: the Gateway rejects the command, its API having no such checkbox.

`RECONNECTDATA`
: Refreshes the market data connections, as Ctrl-Alt-F does. Not valid for the FIX Gateway.

`RECONNECTACCOUNT`
: Reconnects to the IB login server, as Ctrl-Alt-R does. Not valid for the FIX Gateway.

`EXIT`
: Closes the connection to the command server.

For TWS, `RESTART` uses the *File > Restart…* menu command, so the restart starts immediately. The Gateway has no such command: IBC sets the auto-restart time in the *Lock and Exit* section to the start of the next minute (or of the one after, when less than 58 seconds into the current minute) and displays a countdown overlay on the main window. That time remains in force after the restart, so either let `AutoRestartTime` in `config.ini` override it, or send another `RESTART` once the session is back to set it to its usual value.

Any Questions?

The retired upstream project used the support forum at https://groups.io/g/ibcalpha. For this repository, use the issue tracker at https://github.com/hartza-capital/IBC/issues.

When reporting a problem, provide as much evidence as you can: the versions of IBC (the `version` file in the installation) and of TWS/Gateway, a description of the incorrect behaviour, and the IBC log. The log usually identifies the source of a problem quickly.

Changes from IBController

IBC was forked from the original IBController project and differs from it in ways that matter when you switch:

1. The program file is called `IBC.jar`.
2. Settings file changes:
   - the file is called `config.ini`, where IBController used `IBController.ini`;
   - `ForceTwsApiPort` was renamed `OverrideTwsApiPort`;
   - `AcceptIncomingConnectionAction` now defaults to `manual`, so API connections from unknown computers must be accepted explicitly;
   - `PasswordEncrypted` and `FIXPasswordEncrypted` were removed, along with the password 'encryption' facility;
   - `IbControllerPort` was renamed `CommandServerPort`, and defaults to `0`, meaning the command server is not started;

- `IbControlFrom` was renamed `ControlFrom`;
- `IbBindAddress` was renamed `BindAddress`.

3. Script names on Linux:

```
IBControllerStart.sh          ->  twsstart.sh
IBControllerGatewayStart.sh   ->  gatewaystart.sh
```

Building IBC from source

Building requires a Java 25 JDK and the `IBC_BIN` variable pointing at the `jars` folder of an installed TWS/Gateway. The Maven wrapper provides Maven itself.

```
export IBC_BIN=~/Jts/1050/jars
make dist          # dist/IBCLinux-<version>.zip
make test          # JUnit 5 suite
make checkstyle    # style gate
make docker-verify # checks the ZIP as a consumer receives it
```

`CONTRIBUTING.md` describes the build in detail and `docker/README.md` the containerised release checks.

To regenerate this guide after editing `docs/userguide.md`, run `make userguide`, which rebuilds `resources/userguide.pdf`; it needs `pandoc` and `tectonic` on `PATH` (`brew install pandoc tectonic`).

Licence

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